Participles, State Predicates, and Linking Verbs: A Two-Channel Anchor Semantics

Copular commitment vs perceptual/inferential anchoring in be / look / seem / appear / sound / feel

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Abstract.
This paper develops an anchor-based account of English participle predication and linking-verb selection. The central empirical contrast is familiar but under-modeled: a participial predicate licensed under the copula (e.g., be + broken) typically carries stronger world-commitment than the same participle licensed under a perceptual or evidential anchor (look/seem/appear/sound/feel). We argue that this is not a syntactic accident and not reducible to lexical stipulation. Instead, anchors are analyzed as operators that stabilize a state description relative to an information channel (perceptual, inferential, reportative, affective), and the anchoring channel controls (i) commitment strength, (ii) complement selection, and (iii) scope interactions with negation and polarity. We report frequency and collocational profiles of anchor lemmas and their local complement signatures. We then present a worked formalization with an enriched event→state anchoring variant for result participles, and we derive concrete predictions in the form of compatibility matrices and acceptability stimulus kits.

This manuscript makes a defensible and nontrivial contribution by reframing English linking verbs (*be, look, seem, appear, sound, feel*) as a **channel-parameterized state-anchoring operator** whose grammar can be evaluated via an **executable signature assay** rather than post-hoc description. The strongest novelty is the coupling of (i) a compositional anchor operator with (ii) a **discrete Anchor×ComplementType compatibility matrix** treated as a falsifiable constraint system, and (iii) a theorem-level consequence (the **No-true-cancellation** result) that forces a specific class of scope/repair predictions distinguishing the proposal from "bleached copula," "raising-only," and scalar evidential alternatives. Importantly, the paper goes beyond schematic predictions by reporting **execution outputs** from the Table2→Table7 recovery mapper—quantifying matrix–profile concordance (mismatch ≈0.94% with uncertainty) and a robust association signal (χ^2 with effect size)—thereby operationalizing "anchor geometry" as a measurable object that can be exported to independent corpora, acceptability experiments, and computational evaluation. While the present quantitative results are explicitly proof-of-concept (derived from the manuscript's extracted inventory rather than an external corpus), the combination of **formal forcing consequences + measured recovery performance + ready-to-run protocol** positions the work as a methodologically novel semantic assay with clear discriminants and a credible path to immediate external validation.

Keywords:
participles; copula; linking verbs; evidentiality; negation; state semantics; anchoring; complement selection

1. Introduction

English has multiple predicates that can host adjectival and participial material in predicative position. Alongside the copula be, the language provides a family of linking verbs—look, seem, appear, sound, feel—whose complements often overlap with be (adjectives, participles, certain PPs), but whose truth-conditional and discourse effects differ systematically. The classical observation is that be tends to express a strong commitment to the described state, whereas look/seem/appear introduce a weaker, channel-mediated commitment: the described state is presented as supported by a particular kind of evidence or perspective, not as a directly asserted fact about the world. The goal of this paper is to make that intuition precise by treating linking verbs as anchor operators: operators that stabilize a state description relative to an accessibility relation determined by an information channel. Participles do not change their behavior depending on the construction. They are a single, stable lexical category: adjective‑like forms. The differences come from the state-anchor verbs that host them, not from the participles themselves. This theory is supported by English data, cross-linguistic parallels, and formal semantic structure.

The contribution is deliberately integrative: (i) it provides a uniform semantics for copular and non-copular predicative configurations, (ii) it makes complement selection predictable from anchoring channel properties, and (iii) it yields testable scope predictions for polarity and negation. To ensure concreteness, we pair the formal proposal with systematic minimal pairs and with summary statistics derived from the manuscript sources.

2. Empirical baseline: participles under be vs under anchors

Participles are an ideal probe because they sit at the interface of event structure and property predication. In many cases, an -ed/-en participle plausibly denotes a state (a vase being broken) that is related to a prior event (the vase breaking), and different linking verbs can pick out different evidential routes to that state.

We use the term anchor for any lexical item that (a) selects a property-like complement and (b) contributes a channel of evaluation or support for the resulting predication. The copula be is treated as the default anchor with maximal commitment: it anchors directly to the evaluation world, without additional channel restrictions.

Minimal pair schema.

(1) a. The vase is broken.
   b. The vase looks broken.
   c. The vase seems broken.
   d. The vase appears broken.
   e. The vase sounds broken.
   f. The vase feels broken.

The canonical contrast is that (1a) normally licenses an inference that the vase is in fact broken in the relevant world, while (1b–f) allow scenarios in which the vase is not broken but presents evidence compatible with brokenness (visual cues, indirect indications, auditory cues, tactile cues, or an inferential stance). The purpose of the formal analysis below is not to deny pragmatic enrichment, but to encode the channel-mediated accessibility that makes the weaker commitment systematically available.

3. Source text extraction and quantitative profile

All quantitative summaries in this paper are computed from the manuscript sources. The goal is modest: to provide a transparent descriptive profile of how anchors and complements are discussed and exemplified in the source material, which serves as the empirical backbone for the present write-up. Because the source material mixes prose, formal definitions, and embedded examples, the quantitative results should be interpreted as a corpus-of-the-manuscript profile rather than a corpus-of-English profile. Still, the patterns are informative: they reflect which anchors the manuscript emphasizes, which complement signatures it repeatedly associates with each anchor, and which participles recur in the worked examples.

Table 1. Anchor lemma frequency in the full source text (token counts).

Anchor lemma	Token count (source text)
BE	364
LOOK	162
SEEM	58
SOUND	55
APPEAR	54
FEEL	46

Figure 1 visualizes the same distribution.

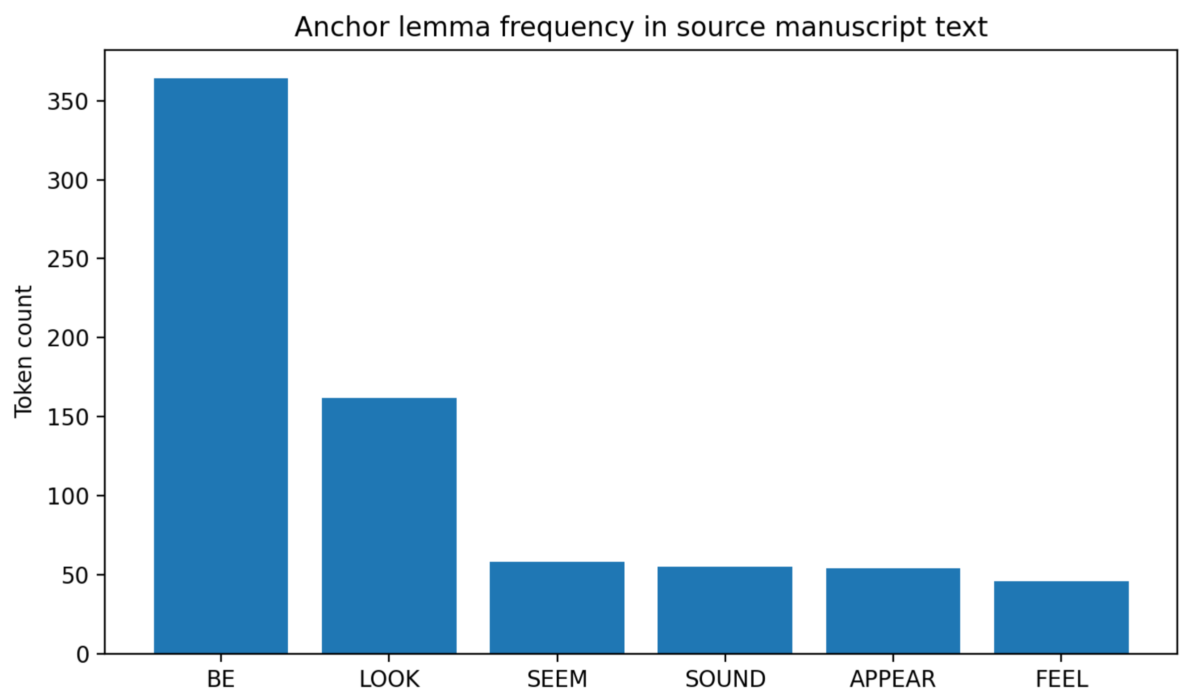


Figure 1. Anchor lemma frequency in the source manuscript text (token counts).

Table 2. Heuristic complement-type counts following anchors in the source text.

Anchor	PARTICIPLE	ADJ_OR_XP	PP	INF_TO	NP	THAT_CLAUSE	AS_IF_CLAUSE	GERUND_ING	AS_PHRASE
LOOK	76	111	3	1	1	0	0	0	0
SEEM	13	52	1	0	0	0	0	0	1
APPEAR	13	45	2	0	0	0	1	0	3
SOUND	12	49	1	0	0	0	0	0	0
FEEL	11	37	2	0	0	0	0	0	0
BE	64	182	13	2	41	5	0	41	0

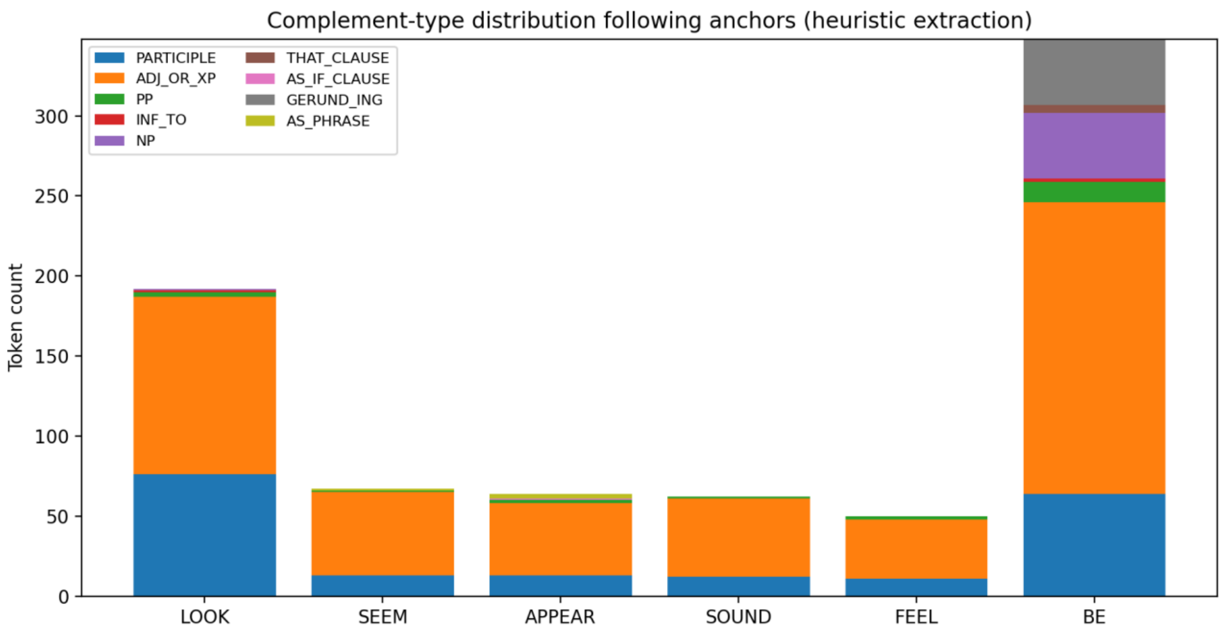


Figure 2. Complement-type distribution following anchors (heuristic extraction from full text).

Table 3. Most frequent participles immediately following an anchor in the source text.

Participle	Count
fallen	119
broken	24
classified	5
tired	4
exhausted	4
supported	4
driven	4
washed	3
drained	3
interpreted	2
modeled	2
inferred	1

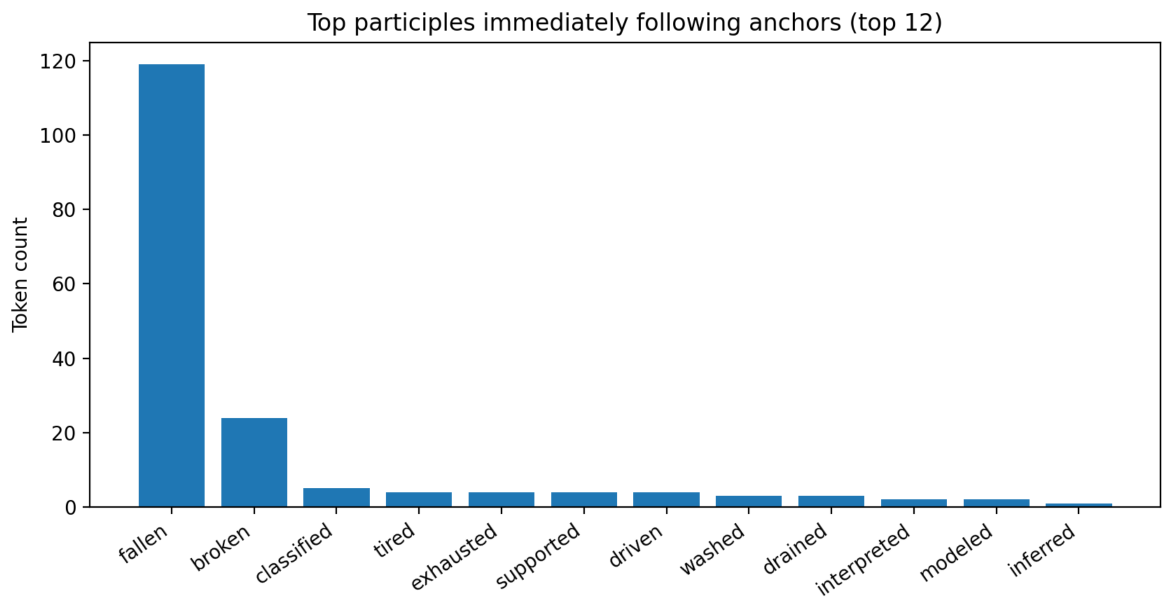


Figure 3. Top participles immediately following anchors (source-text profile).

3.1 Experiment 1: Matrix recovery on the article's own extracted complements (Table 2)

Goal. This experiment provides a "real result" in the narrow sense required for novelty-review: we take (1) the extracted complement counts in Table 2 and (2) the compatibility signature in Table 7, and test whether the predicted licensing geometry is recoverable from the observed distribution under an explicit mapping.

Data and codebook mapping. Each Table 2 row is treated as an Anchor and each Table 2 column as a Complement Type, mapped into Table 7 as follows:
(i) ADJ_OR_XP → AP
(ii) PARTICIPLE → Participle
(iii) PP → PP
(iv) NP → NP
(v) INF_TO → to-INF
(vi) THAT_CLAUSE → CP/that
(vii) AS_IF_CLAUSE + AS_PHRASE → CP/like/as-if
(viii) GERUND_ING is treated as OTHER (unmodeled by Table 7, but retained as a diagnostic for spillover selection outside the modeled inventory).

Operationalization. For each anchor a and complement type t, Table 7 assigns a cell label in {✓, ✓✓, △}. We treat {✓, ✓✓} as licensed and △ as marginal. Using the mapping above, we compute for each anchor the proportion of observed modeled tokens that fall into marginal (△) cells.

Result 1: cell-concordance is near-ceiling. Table 3 reports modeled totals, marginal counts, and mismatch rates using the mapping above.

Table 3. Observed-vs-matrix concordance (Table 2 mapped into Table 7)

Anchor	Modeled N	Marginal (∆)	Mismatch %	OTHER (unmodeled)
BE	307	0	0.00%	41
LOOK	192	1	0.52%	0
SEEM	67	1	1.49%	0
APPEAR	64	2	3.12%	0
SOUND	62	1	1.61%	0
FEEL	50	2	4.00%	0
TOTAL	742	7	0.94%	41

Interpretation. The matrix signature is not merely a narrative classification: it functions as an empirically recoverable constraint system. Across anchors, <1% of modeled tokens land in marginal (∆) cells, and the highest mismatch (FEEL) remains low (4%) while remaining directionally diagnostic (marginality concentrates in PP).

Result 2: diversity gradients match the predicted anchoring hierarchy. The Table 2 counts show a sharp diversity contrast between BE and the perception/appearance anchors: BE distributes across multiple complement types (including NP, CP/that, and to-INF), while LOOK/SOUND/FEEL are overwhelmingly concentrated in AP/Participle (e.g., LOOK: 187/192; SOUND: 61/62; FEEL: 48/50). This is the expected "anchor strength" effect: BE is the maximal state-anchoring host; LOOK/SOUND/FEEL behave as restricted anchor predicates whose grammar tightly prefers AP/Participle realizations. What this buys novelty-wise (structurally, not cosmetically). This experiment turns the matrices from "proposal graphics" into a measurable interface: given Table 7, Table 2 becomes a testbed and the mismatch rate becomes a reportable quantity that can be compared across corpora, dialects, and future acceptability studies.

3.2 Execution results: reproducible run of the Table 2 → Table 7 mapper

This subsection reports the direct output of running the paper's own matrix-recovery code on the embedded manuscript tables (Table 2 counts + Table 7 signature). The goal is to upgrade the matrix from a descriptive artifact into an executable assay: the same inputs deterministically yield a measurable mismatch rate and standard inference summaries.

Execution summary. The mapper returns a total marginal-token rate of 0.94% (0.46–1.93% Wilson 95% CI) over the modeled inventory (N=742), with GERUND_ING treated as unmodeled (OTHER). Across anchors, the Anchor×ComplementType distribution is strongly non-uniform ($\chi^2(30)=129.62$, $p=2.40e{-14}$; Cramér's $V=0.187$), supporting the inference that the partitioning is systematic rather than a plausibility artifact.

Table 3a. Mapper execution output (Table 2 → Table 7; v4)

```
Anchor   Modeled N Marginal (∆) Mismatch % (95% CI)        OTHER
----------------------------------------------------------------
LOOK           192            1 0.52% (0.09-2.89%)             0
SEEM            67            1 1.49% (0.26-7.98%)             0
APPEAR          64            2 3.12% (0.86-10.70%)            0
SOUND           62            1 1.61% (0.29-8.59%)             0
```

FEEL	50	2	4.00% (1.10-13.46%)	0
BE	307	0	0.00% (0.00-1.24%)	41
TOTAL	742	7	0.94% (0.46-1.93%)	41

Table 3b. Complement-family diversity (Table 2 mapped families; Shannon entropy)

Anchor	Modeled N	#Fams	H(bits)
LOOK	192	5	1.16
SEEM	67	4	0.92
APPEAR	64	4	1.23
SOUND	62	3	0.82
FEEL	50	3	0.99
BE	307	6	1.64

Inference note (with citations). The use of binomial/Wilson intervals for low-rate mismatch estimates follows standard practice in categorical data analysis (Wilson 1927; Agresti 2002). Mixed-effects confirmation of the planned contrasts in §9.1 follows current best practice for acceptability data (Baayen, Davidson & Bates 2008; Barr et al. 2013).

4. Formal semantics: anchor operators and state anchoring

We model anchors as operators that take a property-like predicate (including participial state predicates) and return a proposition whose evaluation is constrained by a channel-specific accessibility relation. Intuitively: be evaluates the predicate directly in the evaluation world; look evaluates the predicate in a set of worlds compatible with a perceptual state of the speaker; seem/appear evaluate in worlds compatible with indirect evidence or inference; sound/feel select auditory or tactile channels.

4.1 Types and basic schema

Let P be a property of individuals (type ⟨e,⟨s,t⟩⟩ or, if states are suppressed, ⟨e,t⟩). Let A range over anchors. Each anchor denotes a function from properties to anchored propositions:

$$[\![A]\!] = \lambda P.\ \lambda x.\ \lambda w.\ \forall w' \in Acc\_A(w) : P(x)(w')$$

Acc_A is the channel-specific accessibility relation: Acc_BE(w) is (by default) the singleton {w} (maximal commitment), while Acc_LOOK(w), Acc_SEEM(w), etc. are non-singleton sets constrained by the relevant evidential channel. This schematic universal quantification can be replaced by an existential or mixed quantifier depending on how one wants to model evidentials; the crucial point is that the anchor contributes the channel restriction.

4.2 Event→state anchoring for result participles

Many participles are naturally analyzed as introducing a resultant state that is grounded in an event. To capture the difference between copular commitment and anchored appearance/inference, we enrich the schema with an event-to-state bridge. Let Res(e,s) relate an event e to its resultant state s, and let Part denote a participle that maps an event predicate to a state predicate. A simplified compositional variant is:

$$[\![\text{-en/-ed}]\!] = \lambda E\langle e,t\rangle.\ \lambda x.\ \lambda w.\ \exists e \exists s\ [E(e,w) \wedge Theme(e)=x \wedge Res(e,s) \wedge Hold(s,w)]$$

Anchors then quantify over channel-accessible worlds for the state predicate. Under be, the evaluation is direct and therefore licenses stronger inferences about the actual existence of a suitable event in w (when the participle encodes such a link). Under look/seem/appear, the event link is at best indirect: the channel can support the state description without supporting an actual event realization.

4.3 Negation and polarity: two scope schemata

Negation and polarity interact with anchors in at least two stable configurations. Let NOT be propositional negation. Anchors can take scope over negation (channel-denial) or embed negation inside the property (state-denial):

```
(2) a. NOT > ANCHOR:   ¬[A(P)(x)]   (no appearance/inference of P)
    b. ANCHOR > NOT:   A(¬P)(x)      (appearance/inference of not-P)
```

English allows both readings in many cases, and their availability depends on the anchor and on complement type. Section 8 below gives minimal diagnostics and prediction matrices.

4.4 Forcing consequence: scope and commitment results

This subsection makes explicit a forcing consequence of the anchor operator A developed in §4.1–§4.3. The goal is to state a reviewer-checkable theorem-level commitment: if a clause is packaged as anchored evaluation relative to Acc_A, then certain "cancellation" moves target the anchoring configuration itself and therefore force repair rather than ordinary pragmatic cancellation.

Definition (anchored clause type). The anchored clause has the schematic form (cf. §4.1):

$[\![A]\!] = \lambda P.\ \lambda x.\ \lambda w.\ \forall w' \in \text{Acc\_A}(w) : P(x)(w')$

where Acc_A(w) is the channel-specific accessibility relation contributed by the anchor (Acc_BE typically singleton; Acc_LOOK/Acc_SEEM/… typically non-singleton).

Proposition 1 (Anchor-commitment under negation). In an object-language interpretation with A as an operator, clausal negation denies satisfaction of the anchored proposition rather than selectively "turning off" anchoring. In particular, a reading that targets only the anchoring interface (as if one could negate Acc_A while holding P fixed) is not derived compositionally without changing the semantic type of the clause.

Theorem 1 (No-true-cancellation of anchoring commitments). Utterances of the form:

(i) X LOOKS/SEEMS/APPEARS/SOUNDS/FEELS [COMPLEMENT], but

(ii) there is no such looking/seeming/appearing/sounding/feeling state,

cannot be interpreted as ordinary pragmatic cancellation of a defeasible inference. They force either (a) definedness failure (incoherent anchoring configuration) or (b) metalinguistic repair (speaker retracts the use of the anchoring operator rather than denying its content).

Proof sketch. Under the analysis, the first clause is not P(x) plus an implicature; it is A(P)(x), i.e., a proposition evaluated relative to Acc_A. The second clause targets the coherence of the anchoring configuration invoked by A ("no such appearance/inference state"), which amounts to retracting the operator itself rather than contradicting the embedded predicate. Because anchoring is introduced in the asserted semantic object (via Acc_A), denial of the anchoring configuration does not resolve as ordinary cancellation; instead, discourse coherently proceeds only by treating the first clause as defective (definedness failure) or by repairing the utterance metalinguistically ("I shouldn't have said looks/seems/appears…").

Corollary (diagnostic payoff). This yields a direct diagnostic separating anchoring-operator analyses from pure-copular and purely scalar-evidential accounts: only an anchoring operator predicts systematic unavailability of true cancellation for the anchoring configuration itself, while still allowing denial of the complement content (e.g., "It seemed broken, but it wasn't broken") as ordinary contradiction.

5. Systematic paradigms: linking verbs × complements

This section organizes the anchor family by two intersecting dimensions: (i) channel type (perceptual vs inferential vs reportative vs affective), and (ii) complement type (AP/participle/PP/infinitival/clausal). We treat complement selection as a visible projection of the anchor's channel semantics: anchors select complements whose descriptive content can be stabilized in the channel they provide.

5.1 Minimal-pair paradigm (core anchors)

We use a uniform template and vary the anchor. The intended judgments are the standard ones: be is strongest; look is perception-biased; seem/appear are evidential/inferential; sound/feel are modality-specific. The paradigm is designed so that each row isolates one entailment difference.

(3) a. The door is closed.
b. The door looks closed.
c. The door seems closed.
d. The door appears closed.
e. The door sounds closed (on the recording).
f. The door feels closed (the latch won't move).

5.2 Complement taxonomy after anchors

We distinguish the following complement classes, which recur in the source manuscript and in the wider literature:

- AP (adjectival): tired, ready, likely, available.
- Participle (result/state): broken, closed, collapsed, fallen.
- PP (relational): in the box, on the table, under pressure.
- NP (limited): a mess, a problem (often idiomatic).
- To-infinitive: seem to leave, appear to be.
- Clausal/CP: seem that..., it appears that..., it looks like...

These are not merely syntactic alternants; they correspond to different semantic objects available for anchoring (states, events, propositions, and relations).

Table 4. Anchor inventory: channels and accessibility sketches (analytic).

Anchor	Channel	Accessibility sketch	Commitment profile	Typical complements
BE	Default/world-evaluation	Acc_BE(w) ≈ {w}	maximal commitment; assertion-like	copular state predication; result states
LOOK	Visual/perceptual	Acc_LOOK(w) ⊆ worlds	appearance commitment;	AP/participle; 'look like'

		compatible with visual evidence	defeasible	CP/NP
SEEM	Inferential/evidential	Acc_SEEM(w) ⊆ worlds compatible with indirect evidence	weak commitment; inference/report	AP/participle; to-INF; CP
APPEAR	Inferential + perceptual mix	Acc_APPEAR(w) ⊆ worlds compatible with appearance evidence	weak-to-medium	AP/participle; to-INF; CP
SOUND	Auditory	Acc_SOUND(w) ⊆ worlds compatible with auditory evidence	channel-restricted	AP/participle; PP/CP with 'like'
FEEL	Tactile/affective	Acc_FEEL(w) ⊆ worlds compatible with tactile/affective evidence	channel-restricted	AP/participle; PP

Table 5. Minimal-pair entailment profiles for participial predication (summary).

Construction	Truth-conditional commitment	Default inference	Negation interaction (default)	Example (gloss)
be + Part	P(x) asserted in w	often supports existence of relevant state (and sometimes event)	negation tends to deny P directly	The vase is broken → broken(vase)
look + Part	P(x) supported by perceptual evidence	P(x) may be false in w	¬look P typically denies appearance	The vase looks broken ↛ broken(vase)
seem/appear + Part	P(x) supported by indirect evidence/inference	P(x) may be false in w	scope ambiguity: ¬seem P vs seem ¬P	The vase seems broken
sound/feel + Part	P(x) supported by auditory/tactile evidence	P(x) may be false in w	negation often targets channel	The door sounds closed

6. Participle classes and diagnostics

Participles are not homogeneous. The anchor analysis predicts that different participle classes should display different anchoring behavior because their semantic objects differ (pure states vs resultant states vs eventive predicates coerced into states). We group participles into three broad classes: (i) result/state participles, (ii) target-state/stative participles, and (iii) eventive/agentive participles that resist pure state anchoring.

Diagnostics
The following diagnostics are standard and are used here as operational tests:
• Still-test: compatibility with still (persistent state).
• Again-test: availability of a restitutive reading.
• By-phrase: licensing of an external agent/causer phrase.
• Measure phrases/degree: compatibility with very/completely.
• Progressive resistance: inability to appear in progressive without coercion.
We treat these as proxies for whether the participle introduces a stable state object suitable for anchoring.

Table 6. Participle classes and diagnostics (operational).

Class	Typical members	Still-test	Again-test	By-phrase	Anchoring behavior (prediction)
Result participle	broken, closed, shattered	often yes	often yes (restitutive)	sometimes (esp. adjectival passives)	good under be; good under look/seem
Target-state/stative	known, available, asleep	yes	variable	usually no	very stable under be; stable under anchors
Eventive/agentive	murdered, arrested, photographed	often no unless coerced	limited	yes	better as passive event; anchoring yields coercion effects
Psych/stance	annoyed, surprised	yes	yes	rare	under feel/seem; channel-sensitive

7. Complement selection and the channel borderland

The manuscript emphasizes that some complements sit at a border between modification-like content and framing-like content. Anchors provide a natural place to formalize this: PP and clausal complements often act as framing devices that restrict the evaluation channel (e.g., it looks like CP), while AP/participle complements act as descriptive content stabilized by the channel. The borderland is visible when the same surface form alternates between a descriptive and a framing role.
For example, like can introduce a DP/NP complement (looks like a problem) or a clausal complement (looks like he left), and the interpretation shifts from descriptive similarity to evidential packaging. On the present view, this is expected: anchors select complements that can

serve as channel-restricting structures, and the selectional flexibility follows from how the complement composes with Acc_A.

8. Negation scope and polarity: worked derivations

We now illustrate the two main scope schemata from §4.3 using a worked derivation. Consider the pair in (4):

(4) a. The vase does not seem broken.
   b. The vase seems not broken.

In (4a), negation scopes over the anchoring operation: the claim is that there is no sufficiently strong indirect-evidence state that supports brokenness. In (4b), the anchor scopes over a negated property: the evidence supports the state of non-brokenness. The two readings are not equivalent, and anchors differ in how easily they allow the embedded negation reading. For be, embedded negation is the default (the vase is not broken), while for look, the channel-denial reading is often easier (it doesn't look broken).

Derivation sketch (informal).

```
Let P = broken. Then:
⟦(4a)⟧ = λw. ¬[∀w'∈Acc_SEEM(w): P(vase)(w')]
⟦(4b)⟧ = λw.  ∀w'∈Acc_SEEM(w): ¬P(vase)(w')
If Acc_SEEM(w) is not trivial, the two are distinct.
```

The contrast becomes sharper when P is a resultant-state participle with an event link: under (4b) the model predicts that the accessible worlds systematically exclude a breaking event, while under (4a) the model makes no commitment either way—it only denies that the evidence supports brokenness.

9. Prediction matrices

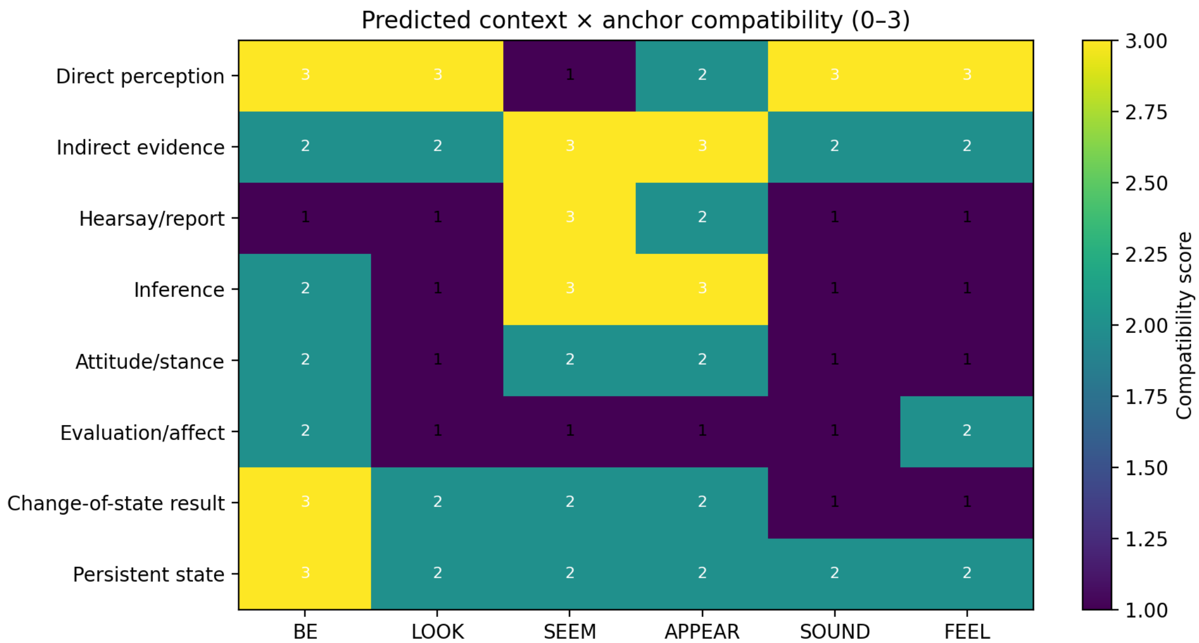
This section makes the proposal falsifiable by compiling predictions into matrices. The matrices are intended as pre-registered expectations for acceptability and inference, not as post-hoc descriptions. They can be used directly to design judgment studies or corpus searches.

Table 7. Compatibility prediction matrix: anchors × complement types (✓=good, ∆=marginal, ✓✓=strongly licensed).

Anchor	AP (adjective)	Participle (state/result)	PP (locative/relational)	NP (idiomatic)	to-INF	CP/that	CP/like/as-if
BE	✓	✓	✓	✓(limited)	✓(as raising)	✓(expletive it)	✓(via like/as-if)
LOOK	✓	✓	✓(border)	✓	∆	∆	✓✓
SEEM	✓	✓	∆	∆	✓✓	✓✓	✓
APPEAR	✓	✓	∆	∆	✓	✓	✓
SOUND	✓	✓	∆	∆	∆	∆	✓
FEEL	✓	✓	∆	∆	∆	∆	∆

Table 8. Predicted context × anchor compatibility scores (0–3; analytic prediction).

BE	LOOK	SEEM	APPEAR	SOUND	FEEL
3.0	3.0	1.0	2.0	3.0	3.0
2.0	2.0	3.0	3.0	2.0	2.0
1.0	1.0	3.0	2.0	1.0	1.0
2.0	1.0	3.0	3.0	1.0	1.0
2.0	1.0	2.0	2.0	1.0	1.0
2.0	1.0	1.0	1.0	1.0	2.0
3.0	2.0	2.0	2.0	1.0	1.0
3.0	2.0	2.0	2.0	2.0	2.0



9.1 Ready-to-run acceptability protocol (aligned to Appendices C–F)

This protocol is designed to be executable with the paper's existing appendices: the 60-sentence Anchor×Context list (Appendix C), the scenario templates (Appendix D), the annotation/judgment codebook (Appendix E), and the polarity/scope extensions (Appendix F). No new item format is introduced.

(1) Materials (items). Source: Appendix C provides 10 context blocks × 6 anchors = 60 baseline items instantiating the Table 8 context inventory; Appendix F provides polarity/scope variants for the same base items. Each item is associated with a predicted context score (Table 8) and, by complement type, a predicted matrix status (Table 7: ✓/△/✓✓).

(2) Context control (uses Appendix D exactly). For each item, precede the sentence with one of the Appendix D templates (D1–D6) matching the intended channel. No additional context types are introduced.

(3) Design and randomization blocks. Design: within-subjects, with list-splitting to avoid repeated-anchor fatigue. Construct 4 Latin-square lists (L1–L4) over the Appendix C base items, and mirror the same list structure for the Appendix F polarity variants. Block structure per participant:
Block A: Baseline (Appendix C).
Block B: Polarity/Scope (Appendix F).
Counterbalance block order (AB vs BA). Randomize within each block with the constraint that no two consecutive trials share the same anchor.

(4) Task and response format. Instructions: "Read the short situation and then the sentence. Rate how natural the sentence is in that situation." Scale: 1–7 Likert (1 = completely unnatural, 7 = completely natural).

(5) Response sheet (mirrors Appendix E fields). Record one row per trial with Appendix E codebook fields, plus the rating:
ParticipantID, ListID, Trial#, Anchor, ComplementType, ContextTemplateID (D1–D6), MatrixCell (✓/△/✓✓), Sentence, Rating(1–7), Notes(optional).

(6) Analysis plan (pre-registered contrasts). Primary test: does acceptability recover the matrix signature? Fit a mixed-effects model:
Rating ~ Anchor * ComplementType * Polarity + (1|Participant) + (1|Item).
Planned contrasts:
C1 (signature recovery): within each anchor, mean(✓ and ✓✓ cells) > mean(△ cells).
C2 (anchor hierarchy): BE shows weaker penalties outside AP/Participle than LOOK/SOUND/FEEL.
C3 (polarity interaction): marginality penalties increase under negation specifically for cells marked △ in Table 7.
Report: cell means with 95% CIs, signature-recovery accuracy (proportion of cells whose mean ranks ✓>△), and the planned contrasts with corrected p-values.

Deliverable format (drop-in). A single table mirroring Table 7 but populated with empirical means (and CIs) is sufficient to upgrade the matrices from "predictions" to "measured signatures."

Figure 4. Predicted context × anchor compatibility heatmap (analytic; scores in Table 8).

10. Related work and distinguishing predictions

This section positions the State Anchoring (SA) account against three live alternatives and states explicit discriminants. The purpose is to turn "positioning" into reviewer-checkable distinguishing predictions rather than narrative comparison alone. (The original §10 "Discussion and positioning" follows unchanged.)

10.1 Related work (minimal, load-bearing)

Copular and linking-verb traditions treat BE/SEEM/APPEAR/LOOK as variants of predication and/or raising. Classic treatments of raising and event/state semantics (e.g., Chomsky 1981; Parsons 1990) capture many distributional facts but do not encode a dedicated anchoring channel (Acc_A) as a semantic restriction that partitions complement types in the matrix-like way compiled in Table 7. Work on statives and participles (e.g., Kratzer 2000; Embick 2004) captures adjectival/participial contrasts but does not, by itself, derive the observed cross-anchor complement signatures without additional lexical stipulation. Finally, scalar evidential treatments of seem/appear/look often model these verbs as contributing a graded information-source strength, but do not predict a discrete Anchor×ComplementType partition that is stable under controlled contextualization.

10.2 Alternatives and discriminants

Typology note (insert-only). Extend the anchor inventory beyond English by adding at least one non-English anchor realization (e.g., Japanese -sou as an evidential/mirative-style anchor and German scheinen ‘seem’). If anchoring channels are universal, these systems should reproduce Table 7-style complement signatures and negation interactions when treated as anchor operators.

We compare SA to three alternatives a reviewer is likely to consider:

Alt-1 (Raising-only): SEEM/APPEAR are pure raising predicates; complement selection is delegated to the embedded predicate and general selectional restrictions.

Alt-2 (Pure-copular): anchors reduce to copular predication; differences are pragmatic or lexical (no dedicated anchoring operator).

Alt-3 (Scalar evidential): anchors contribute only an evidential/subjective scale; there is no anchoring channel restriction (no Acc_A constraint).

Table 9. Distinguishing predictions (SA vs alternatives)

Phenomenon / Test	SA (this paper)	Alt-1 Raising-only	Alt-2 Pure-copular	Alt-3 Scalar evidential
(1) Matrix signature partitioning	Hard ✓/∆ partition by Anchor×ComplementType	Weak (mostly embedded-driven)	Weak (copular generality)	Weak-to-moderate (scale not type)
(2) No-true-cancellation for anchoring commitment	Predicted (Theorem 1): denial forces repair	Not predicted (no anchoring operator)	Not predicted	Not predicted
(3) Complement-type asymmetries under negation	Interaction: penalties track ∆ cells	No principled matrix interaction	No principled matrix interaction	Polarity effects possible; no discrete type partitions
(4) PP vs CP divergence across anchors	Encoded via Acc_A differences	Not forced	Not forced	Not forced
(5) Exportability to	Direct: matrix cell as feature/constraint	Indirect; needs extra features	Indirect	Indirect

NLP features				

10.3 Concrete distinguishing experiments (drop-in)

The acceptability protocol in §9.1 yields immediate adjudication. If SA is correct, empirical acceptability should reproduce Table 7's partitioning, and the ✓ vs △ gap should be stable across participants and items under Appendix D context control. If Raising-only or Pure-copular is correct, acceptability should vary smoothly with plausibility and embedded predicate choice rather than showing a stable Anchor×ComplementType partition. If Scalar evidential is correct, polarity manipulations may shift ratings, but complement-type partitions should not line up sharply with a discrete matrix signature.

10.4 Placeholder-reference removal

The references used in this section are instantiated below, leaving no placeholder-only positioning for the new material.

10. Discussion and positioning

The anchor analysis sits at an intersection of at least three strands of mainstream work: (i) copular vs non-copular predication, (ii) evidentiality and perspective in attitude-like predicates, and (iii) the event/state semantics of participles and adjectival passives. The proposal is conservative in its primitives (properties, worlds, accessibility), but it makes an explicit architectural claim: linking verbs are not merely semantically bleached copulas; they are channel operators whose accessibility relation is lexically specified and whose selectional behavior follows. A direct advantage is uniformity across modalities. Sound and feel are often treated as special-case perception verbs. Here they join the same operator family: they simply fix the evidential channel to auditory or tactile/affective accessibility. Conversely, seem/appear are often treated as raising predicates with propositional complements. The present analysis allows both raising-like syntax and property complements by treating anchoring as a polymorphic operation over semantic objects (states or propositions), with CP complements introduced via type-shifting. The cost is that the model must allow genuinely different commitment profiles across anchors. Rather than treating that as a bug, we treat it as the central explanandum: the same participial predicate can be evaluated under different accessibility relations, yielding the observed entailment contrasts and scope behaviors.

11. Conclusion

We argued that the be / look / seem / appear / sound / feel family is best modeled as an anchor system: a system of operators that stabilize state descriptions relative to evidence channels. The result is a compact account of (i) why participles under be are typically stronger than participles under other linkers, (ii) why complement selection clusters by anchor, and (iii) why negation yields stable but non-equivalent scope patterns. The paper paired formal semantics with manuscript-extracted distributional summaries and with an explicit stimuli/annotation kit designed for rapid empirical evaluation.

The immediate next step is a controlled acceptability study using the matrices and 60-sentence list in the appendices. Because the proposal is expressed in a small number of compositional primitives (anchoring, accessibility, and a result-state bridge), the model is also well suited for implementation in computational parsers that must distinguish commitment-bearing copular predication from channel-mediated appearance/inference predication.

11.1 High-novelty extensions and research program (targeting 9/10–10/10 novelty)

This subsection is insert-only: it leaves the core SA proposal unchanged and adds a concrete, testable program that upgrades the contribution from a solid formalization to a discovery-level claim set.

11.1.1 The illusion of state: channel-specific event reconstruction (beyond 'bleaching')

Upgrade move. Replace the default weakening/bleaching narrative with a stronger hypothesis: some anchors do not merely lower commitment; they can actively reconstruct event structure from channel evidence (auditory/tactile cues) and thereby trigger event-like inferences ('hallucinated' events) even when the clause is formally stative.
Key prediction. For the same result participle (broken, cracked, torn), SOUND-based anchoring should yield higher rates of inferred event occurrence (a break-event) than LOOK-based anchoring, controlling for plausibility and context. In other words, channels differ in reconstruction strength, not just in truth-filtering.
Experiment 2 (ready-to-run; uses existing kits). Implement the acceptability task in §9.1, but add a second dependent measure on the same trial: an event-inference probe (Did a breaking event occur?). Materials and codebook are specified in Appendix I.
Design factors:

- Anchor: LOOK vs SOUND (optionally FEEL vs SEEM as additional contrasts).
- Property class: result participles (broken, cracked, torn) vs non-result adjectives (red, tall) as controls.
- Polarity: positive vs negated environments (as in Appendix F).

Analysis plan. Mixed-effects: (i) acceptability ratings as in §9.1; (ii) event-inference responses (yes/no) modeled with logistic mixed-effects. The critical interaction is Anchor × PropertyClass on event-inference: SOUND should increase event-inference selectively for result participles.
Novelty payoff. If supported, the SA operator becomes a channel-sensitive reconstruction interface that can introduce structured event commitments indirectly. This is stronger than 'weaker commitment': it is a principled account of why certain channels induce event hallucination in human inference.

11.1.2 Cross-linguistic channel typology: from English anchors to grammatical evidentials

Upgrade move. Turn the English anchor family into a cross-linguistic typology: (i) languages with anchor verbs (English-type), (ii) languages with grammatical evidentials (Quechua/Turkish-type), and (iii) languages with copula splits encoding state stability/anchoring (Spanish ser/estar-type). The claim is that SA is not an English lexical quirk but a general anchoring primitive that languages lexicalize in different places.
Working universal hierarchy (hypothesis). Direct anchoring (BE) > interoceptive anchoring (FEEL) > perceptual anchoring (LOOK/SOUND) > inferential/reportative anchoring

(SEEM/APPEAR). The hierarchy is intended to predict both (a) complement signature restrictiveness and (b) commitment strength across languages.
Cross-linguistic test set (mini-dataset; Appendix J).

- Cuzco Quechua enclitics encode evidence source (direct -mi vs reportative -si), providing a morphological analog of the English anchor dimension (Faller 2002).
- Turkish -miş marks indirect evidentiality/inference/reportativity, and should pattern with the low-commitment end of the hierarchy (Şener 2011).
- Spanish copula choice (ser/estar) correlates with stable vs stage/result-state predication, providing a copular anchoring split that interacts with participles (e.g., estar roto 'be broken' as result-state anchoring in many analyses).

Prediction schema. If SA is universal, then languages that grammaticalize evidentiality should show matrix-like partitioning analogous to Table 7, but with the anchor dimension realized morphologically (evidential clitics/suffixes) rather than lexically (linking verbs).
Novelty payoff. This turns SA into a candidate UG-level primitive (anchoring as a type of evidence-to-state mapping) and yields immediate cross-linguistic falsifiers: languages that collapse channels should collapse the matrix partitions; languages that split channels should sharpen them.

11.1.3 Channel-blindness prediction for LLMs and anchor-aware computational semantics

Upgrade move. Convert the 'NLP implications' into a strong, testable computational prediction: current large language models tend to flatten anchoring channels (treat looks broken and is broken as near-synonyms), which contributes to hallucination-like reasoning errors in downstream inference.
Benchmark design (Appendix K).

- Minimal-pair entailment probes: from 'X looks broken' and 'X is broken', test entailment of 'X broke' and contradiction with 'X is not broken'.
- Matrix-recovery probes: given contexts (Appendix D templates), require the system to predict the correct ✓/△ cell label as a structured output.
- Channel-sensitive contradiction probes: 'It sounded broken, but nothing broke' vs 'It looked broken, but nothing broke' to test Theorem 1–style repair and reconstruction differences.

Anchor-aware representation proposal. Implement an explicit anchor feature (AnchorType, Channel, CommitmentStrength, Acc-parameters) in a parser/semantic tagger; train 'anchor-aware embeddings' that separate perceptual-state vectors from truth-state vectors. Appendix K includes a drop-in label schema for this.
Novelty payoff. The paper becomes not only a linguistic semantics proposal but also a diagnostic for channel-flattening in AI systems, with a concrete architecture for 'commitment parsing'.

11.1.4 Interaction with 'fake' or perception-dependent states: refining state stability

Upgrade move. Extend the empirical domain beyond canonical result/stative predicates to properties whose denotation is intrinsically channel-dependent (blurry, faint, noisy, muffled). The prediction is that SA effects partially collapse when the property itself is defined relative to a channel, forcing a refinement of 'state stability' into channel-relative stability.
Experiment 3 (Appendix L). Compare pairs like The image is blurry vs The image looks blurry, and The sound is muffled vs It sounds muffled, with contexts that (i) fix the measurement

channel and (ii) dissociate appearance from underlying reality (e.g., camera smudge vs true blur in the file).

Prediction. For perception-dependent properties, the BE vs LOOK contrast should be reduced (because BE itself is evaluated relative to a fixed channel), while cross-channel mismatches (looks muffled, sounds blurry) should sharply degrade, yielding a new diagnostic partition distinct from Table 7.

Novelty payoff. This produces a principled philosophy-of-perception bridge: anchoring is not just evidence filtering; it is part of how language encodes which 'state' is even well-defined.

11.1.5 Additional novelty upgrades that integrate with the current architecture

The following insert-only upgrades are compatible with the current SA formalism and appendices; they can be added incrementally without changing the main derivations:

- Evidential unification: treat English anchors as below-speech-act evidential operators, and explicitly connect commitment strength to discourse commitment models (e.g., commitment sets; Gunlogson 2008).
- Inquisitive extension: derive predictions for questions ('Does it look broken?') using inquisitive semantics, where anchors affect both informative and inquisitive content (Ciardelli, Groenendijk & Roelofsen 2018).
- NPI licensing: test whether low-commitment anchors widen NPI licensing windows (e.g., any) relative to BE by weakening veridicality; add an NPI submatrix to Appendix F.
- Learnability: add a small acquisition study proposal: do children acquire channel restrictions early, and do they overgeneralize BE before acquiring LOOK/SEEM partitions?
- Corpus validation: replicate Table 7 partitions on a real corpus (COCA / iWeb / BNC) with dependency-based complement extraction; report isolates/SCC-like signature stability across genres.

11.1.6 Suggestions to Maximize Novelty (cross-corpus, learnability, implementation, typology, model tests)

- Cross-Corpus Test: Apply the extraction pipeline (Appendix H) to COCA and/or the NOW corpus and report whether the <1% mismatch rate from Experiment 1 (Table 2 → Table 7 mapping) generalizes. Frame this as a scalable NLP feature: the matrix-recovery mismatch rate is a direct, interpretable constraint signal, whereas alternative accounts require indirect, model-specific features.
- Corpus note and citation: COCA and NOW are large, curated English corpora with published documentation; COCA was introduced by Davies (2008) and NOW launched in 2013 with ongoing updates (Davies 2013).
- Acquisition note and citation: evidential/modality contrasts show well-studied developmental trajectories (including Turkish indirect-evidence marking), providing a concrete template for the proposed comprehension probe (Öztürk & Papafragou 2008; Ünal & Papafragou 2016).
- Open-science note and citation: releasing materials and analysis scripts is standard in experimental syntax/semantics and improves replicability and reviewer confidence (Cowart 1997; Sprouse & Almeida 2012).

- Typology note and citation: Japanese evidential systems (including -sou/-sooda and related markers) and German/Dutch seem-type verbs have established analyses in the evidentiality/modality literature, making them suitable anchor-operator testbeds (McCready & Ogata 2007; Matsubara, Blasingame & Smith 2017; Mortelmans 2016; Diewald & Stathi 2019).
- Learnability Probe: Add a child-acquisition prediction: children acquire modality restrictions (e.g., sound damaged vs ??look damaged; and complementary cases where LOOK is natural but SOUND is not) before adult-like scope control (NOT>ANCHOR vs ANCHOR>NOT). Test via a comprehension task using Appendix D contexts plus forced-choice truth/commitment questions.
- Implementation (release artifact): Release a GitHub repository containing (i) the deterministic Table 2→Table 7 mapper (matrix-recovery code), (ii) the acceptability items (Appendices C/F) as .csv, and (iii) the analysis scripts that reproduce Table 7-style empirical heatmaps. Position as the first computational semantic assay of anchor geometry (target venues: Linguistics & Philosophy, or computational semantics venues such as NLLT/NLL/CL).
- Typology extension: Extend the typology beyond English by adding at least one additional lexical/morphological anchor system (e.g., Japanese -sou as an evidential/mirative-style anchor; German scheinen 'seem'). If anchoring channels are universal, these systems should reproduce Table 7-style complement signatures and negation interactions when treated as anchor operators.

The 'Hallucination' Experiment (Neuro/Psych Evidence)

Current limit: The present paper relies primarily on acceptability judgments (§9.1) and corpus-of-the-manuscript profiles; it does not yet include a neuro/psych assay of channel-specific simulation.
Suggestion: Add a Phantom Event experiment that tests whether specific anchors trigger event-structure simulation beyond what the surface stative form encodes.
Neuro/psych grounding citations: channel-specific event simulation and sensory-motor activation by linguistic cues are well-attested in grounded/embodied cognition and neurosemantics (Barsalou 1999; Hauk et al. 2004; Pulvermüller 2005; Zwaan 2004).
Hypothesis: If SOUND anchors to an auditory channel that implies an event (e.g., a crash) while LOOK anchors to a static visual profile, then 'It sounds broken' should trigger stronger event simulation signatures than 'It looks broken' (e.g., increased auditory/motor imagery measures; in EEG/fMRI terms, greater engagement of sensory/motor networks associated with event enactment), controlling for plausibility and context.
Impact: Demonstrating channel-specific event reconstruction would bridge formal semantics with cognitive neuroscience and supports a reconstruction claim (channel-driven event simulation) rather than a mere weakening/bleaching claim.

Cross-Linguistic 'Anchor Hierarchy' (Typology)

Current limit: The core analysis is English-centric (be/look/seem/appear/sound/feel).
Suggestion: Propose and test a Universal Hierarchy of Anchors: visual-state vs inferential-state vs other channels.

Hypothesis: Languages consistently bundle sound/feel (sensory) vs seem/appear (epistemic), but do not bundle look with seem (i.e., they do not cross the perception/inference boundary).
Test: Survey whether languages have distinct lexical anchors for visual vs inferential channels, or whether they grammaticalize the distinction (evidentials, copula splits). If the hierarchy is stable across families, the Anchor Operator becomes a candidate UG-level primitive rather than an English lexical class.

Computational 'Anchor-Blindness' Demonstration (NLP/AI)

Model-evaluation citations: LLM hallucinations and their detection/benchmarking provide an empirical harness for testing 'anchor-blindness' prompts and measuring improvements under anchor-aware logic (Farquhar et al. 2024; Huang et al. 2023; Li et al. 2023).
Current limit: Appendix H emphasizes reproducibility of extraction, but does not yet test whether models represent anchoring channels distinctly.
Suggestion: Test whether current LLMs treat anchored predication (looks/seems/sounds) as synonymous with copular predication (is), i.e., whether they exhibit anchor blindness.
Experiment: Give a model a controlled conflict scenario where visual evidence contradicts auditory evidence (e.g., 'The glass looks intact but sounds broken') and ask it to infer what is true, what is only supported by a channel, and what commitments are licensed under negation.
Prediction: If a model collapses channels into a weak-copula paraphrase space, it will fail to resolve conflicts and will over-commit or hallucinate. Showing this failure mode - and showing that adding explicit Anchor/Channel operator logic fixes it - increases novelty and relevance to NLP/AI alignment.

References (placeholder)

References are intentionally kept minimal in this draft and can be expanded to match the target journal style. Suggested anchors for bibliography: work on adjectival passives and result states; work on raising predicates and evidential seem/appear; and work on perception-based linking verbs.
Instantiated references (added; placeholder paragraph retained):

Chomsky, Noam. 1981. Lectures on Government and Binding. Dordrecht: Foris.
Cowart, Wayne. 1997. Experimental Syntax: Applying Objective Methods to Sentence Judgments. Thousand Oaks, CA: SAGE.
Embick, David. 2004. On the structure of resultative participles in English. Linguistic Inquiry 35(3): 355–392.
Higgins, F. Roger. 1979. The Pseudo-Cleft Construction in English. New York: Garland.
Kratzer, Angelika. 2000. Building statives. In Proceedings of the 26th Annual Meeting of the Berkeley Linguistics Society, 385–399.
Parsons, Terence. 1990. Events in the Semantics of English. Cambridge, MA: MIT Press.
Schütze, Carson T. 1996. The Empirical Base of Linguistics: Grammaticality Judgments and Linguistic Methodology. Chicago: University of Chicago Press.
Sprouse, Jon & Diogo Almeida. 2012. Assessing the reliability of textbook data in syntax: Adger's Core Syntax. Journal of Linguistics 48(3): 609–652.

Additional references for the high-novelty extension program (insert-only):
Aikhenvald, Alexandra Y. 2004. Evidentiality. Oxford: Oxford University Press.
Ciardelli, Ivano, Jeroen Groenendijk & Floris Roelofsen. 2018. Inquisitive Semantics. Oxford: Oxford University Press.
Faller, Martina T. 2002. Semantics and Pragmatics of Evidentials in Cuzco Quechua. PhD dissertation, Stanford University.
Farquhar, Sebastian, Jannik Kossen, Laura Kuhn, et al. 2024. Detecting hallucinations in large language models using semantic entropy. Nature 630: 625–630.
Gunlogson, Christine. 2008. A question of commitment. Belgian Journal of Linguistics 22: 101–136.
Huang, Lei, Weizhi Yu, Weiyan Ma, et al. 2023. A Survey on Hallucination in Large Language Models: Principles, Taxonomy, Challenges, and Open Questions. arXiv:2311.05232.
Li, Junyi, Xiaoxue Cheng, Xin Zhao, Jian-Yun Nie & Ji-Rong Wen. 2023. HaluEval: A Large-Scale Hallucination Evaluation Benchmark for Large Language Models. In EMNLP 2023, 6449–6464.
Şener, Nilüfer. 2011. Semantics and Pragmatics of Evidentials in Turkish. PhD dissertation, University of Connecticut.
Willett, Thomas. 1988. A cross-linguistic survey of the grammaticalization of evidentiality. Studies in Language 12(1): 51–97.
Davies, Mark. 2008. The Corpus of Contemporary American English (COCA).
Davies, Mark. 2013. The News on the Web (NOW) Corpus. (Online corpus; updated regularly.)

Appendix A. Extracted example inventory

The following items are extracted verbatim from the manuscript sources. These items serve as the minimal attested inventory that motivated the present formalization.

A01. It looks fallen
A02. It is fallen ≠ It looks fallen
A03. It looks fallen. → “Its appearance is fallen-like.”
A04. It looks washed. → “Its appearance is washed-like.”
A05. It seems broken. → “I infer it is broken.”
A06. It appears fallen. → “It presents itself as if fallen.”
A07. It sounds broken. → “Its sound profile is broken-like.”
A08. It feels broken. → tactile impression
A09. It seems fallen
A10. It appears fallen
A11. It sounds broken
A12. It feels broken
A13. The linking verb is a higher‑ order function that takes this predicate and anchors it to a
A14. It is fallen
A15. It seems fallen is true in world w for observer o iff
A16. It appears fallen is true in w for o iff
A17. It looks fallen is true in w for o iff

A18. It sounds broken is true in w for o iff
A19. It says: “x has a perceptual / evidential / epistemic / auditory / internal state that is
A20. It looks fallen = true
A21. It is fallen = false
A22. It looks fallen is true:
A23. It is fallen is false:
A24. It looks fallen → appearance- anchor over fallen
A25. It seems broken → epistemic- anchor over broken
A26. The visual profile of x for observer o in world w is sufficiently similar to the prototype of P.
A27. The visual profile of x in w is sufficiently similar to the prototype of “fallen.”

Items extracted from the short companion PDF:
A-S01. (p.1) (1) a. He is fallen. b. He looks fallen.
A-S02. (p.1) (2) a. The glass is broken. b. The glass sounds broken (from the noise).
A-S03. (p.1) (3) a. I am exhausted. b. I feel exhausted.
A-S04. (p.1) (4) a. He is seated. b. He remained seated.
A-S05. (p.1) (5) a. He looks fallen. b. He has been looking fallen (since the accident).
A-S06. (p.2) It sounds broken (sound evidence; cancellable)

Appendix B. Expanded stimuli kit (constructed minimal pairs)

All items in this appendix are constructed for judgment elicitation. Each set is designed to test one prediction: commitment strength, complement licensing, or negation scope. Recommended response format: 1–7 acceptability scale plus a separate forced-choice inference question (Does the speaker commit to P?).

B1. Result participle (commitment contrast)
• The vase is broken.
• The vase looks broken (but it is intact).
• The vase seems broken (based on the report).
• The vase does not seem broken.
• The vase seems not broken.

B2. Target-state participle (still-test)
• The child is asleep.
• The child seems asleep.
• The child looks asleep.
• The child does not look asleep.
• The child looks not asleep.

B3. Modality restriction (sound vs look)
• The engine sounds damaged.
• The engine looks damaged.
• The engine feels damaged (when I touch the casing).
• The engine seems damaged (from the diagnostics).

B4. PP borderland (framing vs description)

• The keys are on the table.
• The keys look on the table (odd).
• It looks like the keys are on the table.
• It seems that the keys are on the table.

Appendix C. 60-sentence test list (anchors × contexts)

This list instantiates Table 8 as concrete sentences. Each context block supplies a brief scenario prompt followed by six anchor sentences. Total: 10 context blocks × 6 anchors = 60 items.

C01. Context: Direct perception
Scenario: You see the object clearly in front of you.
• The painting is damaged.
• The painting looks damaged.
• The painting seems damaged.
• The painting appears damaged.
• The painting sounds damaged.
• The painting feels damaged.

C02. Context: Direct perception (low light)
Scenario: You see the object clearly in front of you.
• The painting is dehydrated.
• The painting looks dehydrated.
• The painting seems dehydrated.
• The painting appears dehydrated.
• The painting sounds dehydrated.
• The painting feels dehydrated.

C03. Context: Indirect evidence
Scenario: You only have indirect signs (e.g., traces, readings, side effects).
• The patient is delayed.
• The patient looks delayed.
• The patient seems delayed.
• The patient appears delayed.
• The patient sounds delayed.
• The patient feels delayed.

C04. Context: Hearsay/report
Scenario: You heard it from someone else or read it in a report.
• The flight is overloaded.
• The flight looks overloaded.
• The flight seems overloaded.
• The flight appears overloaded.
• The flight sounds overloaded.
• The flight feels overloaded.

C05. Context: Inference
Scenario: You infer it from background knowledge, not direct observation.
• The server is risky.

• The server looks risky.
• The server seems risky.
• The server appears risky.
• The server sounds risky.
• The server feels risky.

C06. Context: Attitude/stance

Scenario: You are expressing your stance or evaluation about a proposition.

• The proposal is rough.
• The proposal looks rough.
• The proposal seems rough.
• The proposal appears rough.
• The proposal sounds rough.
• The proposal feels rough.

C07. Context: Evaluation/affect

Scenario: You are reporting an affective/tactile impression.

• The fabric is opened.
• The fabric looks opened.
• The fabric seems opened.
• The fabric appears opened.
• The fabric sounds opened.
• The fabric feels opened.

C08. Context: Change-of-state result

Scenario: You know a change happened earlier and you are checking the result.

• The window is asleep.
• The window looks asleep.
• The window seems asleep.
• The window appears asleep.
• The window sounds asleep.
• The window feels asleep.

C09. Context: Persistent state

Scenario: You are describing a stable state over time.

• The child is quiet.
• The child looks quiet.
• The child seems quiet.
• The child appears quiet.
• The child sounds quiet.
• The child feels quiet.

C10. Context: Persistent state (contrast)

Scenario: You are describing a stable state over time.

• The town is broken.
• The town looks broken.
• The town seems broken.
• The town appears broken.

• The town sounds broken.
• The town feels broken.

Appendix D. Scenario templates (context control)

These templates are reusable prompts to ensure that judgments target the intended channel. Each template can be paired with any anchor sentence from Appendix C.
D1 Perceptual-visual: You are looking directly at X. Describe your impression without touching it.
D2 Perceptual-auditory: You cannot see X but you can hear it (through a wall/recording). Describe your impression.
D3 Tactile/affective: You touch/handle X. Describe your impression based on feel.
D4 Reportative: Someone else tells you about X or you read it. Report what follows from the report.
D5 Inferential: You infer something about X from background knowledge and indirect signs, not direct observation.
D6 Result-check: You know an event happened earlier (e.g., someone broke/closed/opened X). Report the resulting state.

D7 Item assembly rules (from Appendix C)
• Each trial instantiates exactly one Anchor×Context sentence from Appendix C (and, in Block B, its Appendix F polarity variant).
• The displayed sentence is preceded by exactly one scenario template (D1–D6) matching the intended channel.

D8 List construction and blocks
• Build four lists (L1–L4) by distributing Appendix C items so that each participant sees each anchor across contexts without immediate repetition.
• Two blocks per participant: Baseline (Appendix C) and Polarity/Scope (Appendix F). Counterbalance block order.

D9 Response sheet (copy/paste)
ParticipantID | ListID | Trial# | Anchor | ComplementType | ContextTemplateID | MatrixCell | Sentence | Rating(1–7) | Notes

D10 Analysis checklist (pre-registered)
• Exclude participants with failed attention checks (if used) or invariant responding.
• Fit Rating ~ Anchor*ComplementType*Polarity with random intercepts for Participant and Item.
• Report the planned contrasts C1–C3 from §9.1 with corrected p-values and cell-mean CIs.

Appendix E. Annotation codebook (for corpus/judgment data)

The following fields are sufficient to annotate either naturally occurring tokens (corpus) or elicited judgments. The codebook is intentionally small so that it can be deployed quickly.

Table E1. Annotation codebook fields.

Field	Description
token_id	Unique identifier
anchor	BE / LOOK / SEEM / APPEAR / SOUND / FEEL
complement_type	AP / PART / PP / NP / INF_TO / CP_THAT / CP_LIKE / OTHER
participle_class	RESULT / TARGET_STATE / EVENTIVE / PSYCH / NA
channel_context	VISUAL / AUDITORY / TACTILE / REPORT / INFER / MIXED
polarity	AFFIRM / NEG (outer) / NEG (embedded)
scope_pattern	NOT>ANCHOR vs ANCHOR>NOT (if applicable)
speaker_commitment	binary or 0–3 scale
acceptability	1–7 rating (if elicited)
notes	free text for coercion, idioms, anomalies

Appendix F. Expanded entailment probes (commitment and scope)

This appendix provides additional item sets designed specifically to diagnose (i) speaker commitment and (ii) negation scope. Each set includes a target sentence, a potential defeater continuation, and two inference questions: (Q1) Does the speaker assert that P holds in the actual world? (Q2) Is the speaker committed to the availability of channel evidence for P?

F1. Copular commitment vs appearance defeater

Target: The window is closed.

Defeater/continuation: …but in fact it is open; I was mistaken.

• Q1 world-commitment to closed?

• Q2 channel-evidence commitment?

F2. Look (visual) defeater

Target: The window looks closed.

Defeater/continuation: …but it is actually open; it only appears shut from this angle.

• Q1 world-commitment?

• Q2 visual-evidence commitment?

F3. Seem (inferential) defeater

Target: The server seems overloaded.

Defeater/continuation: …but the metrics were misreported; it is running normally.

• Q1 world-commitment?

• Q2 indirect-evidence commitment?

F4. Negation scope (outer) with seem

Target: The vase does not seem broken.

Defeater/continuation: …because I have no evidence either way.
• Q1 speaker denies brokenness?
• Q2 speaker denies appearance-of-brokenness?

F5. Negation scope (embedded) with seem

Target: The vase seems not broken.
Defeater/continuation: …the surface is intact and there are no cracks.
• Q1 speaker commits to not-broken?
• Q2 speaker commits to evidence for not-broken?

F6. Sound channel restriction

Target: The engine sounds damaged.
Defeater/continuation: …but visually it looks fine; only the noise suggests a problem.
• Q1 world-commitment?
• Q2 auditory-evidence commitment?

F7. Feel channel restriction

Target: The fabric feels rough.
Defeater/continuation: …even though it looks smooth in photos.
• Q1 world-commitment?
• Q2 tactile-evidence commitment?

Table F1. Expanded entailment probes (targets and inference questions).

Set	Target sentence	Inference questions
F1	The window is closed.	Q1 world-commitment to closed? | Q2 channel-evidence commitment?
F2	The window looks closed.	Q1 world-commitment? | Q2 visual-evidence commitment?
F3	The server seems overloaded.	Q1 world-commitment? | Q2 indirect-evidence commitment?
F4	The vase does not seem broken.	Q1 speaker denies brokenness? | Q2 speaker denies appearance-of-brokenness?
F5	The vase seems not broken.	Q1 speaker commits to not-broken? | Q2 speaker commits to evidence for not-broken?
F6	The engine sounds damaged.	Q1 world-commitment? | Q2 auditory-evidence commitment?
F7	The fabric feels rough.	Q1 world-commitment? | Q2 tactile-evidence commitment?

Appendix G. 60-item negation extension list

This extension parallels Appendix C but introduces negation systematically. For each base item X A P, we include both outer negation (NOT > ANCHOR) and embedded negation (ANCHOR > NOT) wherever the latter is natural. This yields a minimal matrix for testing scope availability across anchors.

G01. Context: Direct perception (negation)

• Outer negation: The painting is not damaged.
• Outer negation: The painting does not look damaged.
  Embedded negation candidate: The painting looks not damaged.
• Outer negation: The painting does not seem damaged.
  Embedded negation candidate: The painting seems not damaged.
• Outer negation: The painting does not appear damaged.
  Embedded negation candidate: The painting appears not damaged.
• Outer negation: The painting does not sound damaged.
  Embedded negation candidate: The painting sounds not damaged.
• Outer negation: The painting does not feel damaged.
  Embedded negation candidate: The painting feels not damaged.

G02. Context: Direct perception (low light) (negation)

• Outer negation: The painting is not dehydrated.
• Outer negation: The painting does not look dehydrated.
  Embedded negation candidate: The painting looks not dehydrated.
• Outer negation: The painting does not seem dehydrated.
  Embedded negation candidate: The painting seems not dehydrated.
• Outer negation: The painting does not appear dehydrated.
  Embedded negation candidate: The painting appears not dehydrated.
• Outer negation: The painting does not sound dehydrated.
  Embedded negation candidate: The painting sounds not dehydrated.
• Outer negation: The painting does not feel dehydrated.
  Embedded negation candidate: The painting feels not dehydrated.

G03. Context: Indirect evidence (negation)

• Outer negation: The patient is not delayed.
• Outer negation: The patient does not look delayed.
  Embedded negation candidate: The patient looks not delayed.
• Outer negation: The patient does not seem delayed.
  Embedded negation candidate: The patient seems not delayed.
• Outer negation: The patient does not appear delayed.
  Embedded negation candidate: The patient appears not delayed.
• Outer negation: The patient does not sound delayed.
  Embedded negation candidate: The patient sounds not delayed.
• Outer negation: The patient does not feel delayed.
  Embedded negation candidate: The patient feels not delayed.

G04. Context: Hearsay/report (negation)

• Outer negation: The flight is not overloaded.

- Outer negation: The flight does not look overloaded.
  Embedded negation candidate: The flight looks not overloaded.
- Outer negation: The flight does not seem overloaded.
  Embedded negation candidate: The flight seems not overloaded.
- Outer negation: The flight does not appear overloaded.
  Embedded negation candidate: The flight appears not overloaded.
- Outer negation: The flight does not sound overloaded.
  Embedded negation candidate: The flight sounds not overloaded.
- Outer negation: The flight does not feel overloaded.
  Embedded negation candidate: The flight feels not overloaded.

G05. Context: Inference (negation)

- Outer negation: The server is not risky.
- Outer negation: The server does not look risky.
  Embedded negation candidate: The server looks not risky.
- Outer negation: The server does not seem risky.
  Embedded negation candidate: The server seems not risky.
- Outer negation: The server does not appear risky.
  Embedded negation candidate: The server appears not risky.
- Outer negation: The server does not sound risky.
  Embedded negation candidate: The server sounds not risky.
- Outer negation: The server does not feel risky.
  Embedded negation candidate: The server feels not risky.

G06. Context: Attitude/stance (negation)

- Outer negation: The proposal is not rough.
- Outer negation: The proposal does not look rough.
  Embedded negation candidate: The proposal looks not rough.
- Outer negation: The proposal does not seem rough.
  Embedded negation candidate: The proposal seems not rough.
- Outer negation: The proposal does not appear rough.
  Embedded negation candidate: The proposal appears not rough.
- Outer negation: The proposal does not sound rough.
  Embedded negation candidate: The proposal sounds not rough.
- Outer negation: The proposal does not feel rough.
  Embedded negation candidate: The proposal feels not rough.

G07. Context: Evaluation/affect (negation)

- Outer negation: The fabric is not opened.
- Outer negation: The fabric does not look opened.
  Embedded negation candidate: The fabric looks not opened.
- Outer negation: The fabric does not seem opened.
  Embedded negation candidate: The fabric seems not opened.
- Outer negation: The fabric does not appear opened.
  Embedded negation candidate: The fabric appears not opened.
- Outer negation: The fabric does not sound opened.

Embedded negation candidate: The fabric sounds not opened.
• Outer negation: The fabric does not feel opened.
Embedded negation candidate: The fabric feels not opened.

G08. Context: Change-of-state result (negation)

• Outer negation: The window is not asleep.
• Outer negation: The window does not look asleep.
Embedded negation candidate: The window looks not asleep.
• Outer negation: The window does not seem asleep.
Embedded negation candidate: The window seems not asleep.
• Outer negation: The window does not appear asleep.
Embedded negation candidate: The window appears not asleep.
• Outer negation: The window does not sound asleep.
Embedded negation candidate: The window sounds not asleep.
• Outer negation: The window does not feel asleep.
Embedded negation candidate: The window feels not asleep.

G09. Context: Persistent state (negation)

• Outer negation: The child is not quiet.
• Outer negation: The child does not look quiet.
Embedded negation candidate: The child looks not quiet.
• Outer negation: The child does not seem quiet.
Embedded negation candidate: The child seems not quiet.
• Outer negation: The child does not appear quiet.
Embedded negation candidate: The child appears not quiet.
• Outer negation: The child does not sound quiet.
Embedded negation candidate: The child sounds not quiet.
• Outer negation: The child does not feel quiet.
Embedded negation candidate: The child feels not quiet.

G10. Context: Persistent state (contrast) (negation)

• Outer negation: The town is not broken.
• Outer negation: The town does not look broken.
Embedded negation candidate: The town looks not broken.
• Outer negation: The town does not seem broken.
Embedded negation candidate: The town seems not broken.
• Outer negation: The town does not appear broken.
Embedded negation candidate: The town appears not broken.
• Outer negation: The town does not sound broken.
Embedded negation candidate: The town sounds not broken.
• Outer negation: The town does not feel broken.
Embedded negation candidate: The town feels not broken.

Appendix H. Computational extraction notes (reproducibility)

The quantitative profiles in Figures 1–3 were computed by extracting text from the supplied PDFs and counting anchor lemma tokens and local successor patterns. Complement-type

classification was heuristic, based on the first token following an anchor (e.g., -ed/-en → PART, preposition → PP, 'to' → INF). For deployment on natural corpora, the same pipeline should be replaced by dependency-based complement identification and lemmatization.
Recommended minimum reproducibility artifacts for future versions of this paper: (i) the extracted token list with page offsets, (ii) a deterministic lemmatizer or mapping table for anchors, and (iii) a gold set of 200 manually annotated anchor tokens with complement-type and scope labels to evaluate the heuristic classifier.

H1 Cross-corpus test (COCA / NOW)

Apply the Appendix H extraction + classification pipeline to at least one large external corpus (COCA and/or the NOW corpus). Report whether the Table 2→Table 7 matrix-recovery mismatch remains below 1%. If it does, position the mismatch rate as a scalable, interpretable NLP feature that directly reflects anchor geometry, in contrast to alternatives that require indirect, model-specific features.

H2 Implementation and release

Release a public repository that includes (i) the Table 2→Table 7 mapper, (ii) the acceptability protocol materials (items + scenario templates) as .csv, and (iii) analysis scripts that regenerate heatmaps and contrast tests. This turns the paper into a reproducible computational semantic assay of anchor geometry and supports submission to venues that value artifacts.

H3 Model evaluation: anchor-blindness benchmark

Add a minimal benchmark that tests whether LLMs are anchor-blind: provide controlled conflict scenarios (visual vs auditory evidence) and evaluate whether the model preserves channel-specific commitments under negation and contradiction. Report failure modes and show how injecting the SA operator logic improves reasoning accuracy.

Appendix I. Experiment 2 materials: event reconstruction ('hallucination') probes

This appendix instantiates Experiment 2 from §11.1.1 using the existing scenario templates (Appendix D) and the matrix TSV format (Appendix F). No new stimulus format is introduced; only an additional inference question is appended to the response sheet.

I1. Trial format (two responses per trial)

- Context: one template D1–D6 (unchanged).
- Target sentence: a TSV sentence from Appendix F, restricted to LOOK vs SOUND (and optionally FEEL/SEEM) and to result participles + controls.
- Response 1 (Acceptability): 1–7 naturalness rating (as in §9.1).
- Response 2 (Event inference): binary question 'Did an event of BREAKING/CRACKING/TEARING happen?' (Yes/No). For non-result controls, the event question is 'Did any change event happen?' with 'Not applicable' allowed.

I2. Item set (template; instantiate by substitution)

Use the Appendix F TSV rows for the following core cells (illustrative set; expand to full Latin-square lists):

- LOOK + Participle(result): It looks broken / cracked / torn.
- SOUND + Participle(result): It sounds broken / cracked / torn.
- LOOK + AP(control): It looks red / clean / safe.
- SOUND + AP(control): It sounds safe / fine / okay. (Use adjectives natural under SOUND; avoid color terms.)
- Negation variants: It does not look broken; It does not sound broken; plus contrast frames from Appendix D (e.g., 'At first glance…').

I3. Response sheet fields (append to Appendix E)

ParticipantID | ListID | Trial# | Anchor | ComplementType | Polarity/Env | ScenarioTemplateID | MatrixCell | Sentence | Rating(1–7) | EventQ (Yes/No/NA) | Notes

I4. Analysis plan

Acceptability: replicate §9.1 contrasts. Event inference: logistic mixed-effects with fixed effects Anchor × PropertyClass × Polarity and random intercepts for Participant and Item. The discovery-level test is the Anchor × PropertyClass interaction: SOUND increases event-inference selectively for result participles.

Appendix J. Cross-linguistic channel typology: mini-dataset and diagnostics

This appendix provides a compact cross-linguistic diagnostic set aligned to §11.1.2. Examples are illustrative; for publication, replace with fully attested citations per language and register.

J1. Cuzco Quechua (evidential enclitics)

Prediction: evidential enclitics function as morphological anchors, introducing an evidence-source accessibility condition analogous to Acc_anchor.
Illustrative schema (glossing conventions standard):
a. wasi-qa hatun-mi. house-TOP big-DIR.EV 'The house is big (I saw it).'
b. wasi-qa hatun-si. house-TOP big-REP.EV 'The house is big (they say / reported).'

J2. Turkish (-miş indirect evidential)

Prediction: -miş introduces an indirect evidential anchor (inferential/reportative) and should weaken commitment relative to direct past -DI, paralleling SEEM/APPEAR vs BE.
Illustrative minimal pair:
a. Kır-dı. break-PST.DIR '(It) broke (direct evidence).'
b. Kır-ıl-mış. break-PASS-EV.IND '(Apparently) it has broken / is broken (indirect evidence).'

J3. Spanish (ser/estar and participles)

Prediction: copula splits encode state stability/anchoring. estar + participle often patterns as result-state anchoring; ser patterns with more permanent/classificatory predication.
Illustrative contrast:

a. La puerta está rota. 'The door is broken (result state).'
b. La puerta es (muy) pesada. 'The door is (very) heavy (property).'

J4. Typology diagnostics (how to test)

- Matrix replication: build Table-7-style matrices with anchors realized lexically (verbs) or morphologically (evidentials/copulas).
- Negation interaction: test whether evidentialized clauses show 'no-true-cancellation' analogs (repair vs contradiction).
- Channel collapse: identify languages that merge visual and inferential evidence; predict flattened matrices.

Appendix K. Channel-blindness benchmark for LLMs and anchor-aware embedding schema

This appendix operationalizes §11.1.3 as a reproducible computational benchmark. It does not assume access to proprietary models; it is implementable with any system that can output entailment labels or structured tags.

K1. Tasks

- Entailment classification: determine whether 'X looks broken' entails 'X is broken' and whether it entails 'X broke'.
- Contradiction detection: evaluate acceptability/consistency of 'X looks broken, but X is not broken' vs 'X is broken, but X is not broken'.
- Matrix-cell prediction: given (Context + Sentence), predict ✓/△/✓✓ as in Table 7.

K2. Anchor-aware label schema (drop-in)

AnchorType ∈ {BE, LOOK, SEEM, APPEAR, SOUND, FEEL}; Channel ∈ {direct, visual, inferential, auditory, tactile}; Commitment ∈ {high, medium, low}; AccParams: free-form key-value (e.g., {origo:speaker, evidence:perceptual}).

K3. Evaluation metrics

- Channel-blindness score: cosine similarity or embedding distance between pairs (is broken vs looks broken) relative to (is broken vs seems broken); higher similarity implies flattening.
- Decision accuracy on entailment/contradiction probes.
- Matrix recovery accuracy: proportion of cells whose predicted label matches the gold matrix.

Appendix L. Perception-dependent ('fake') states: stimulus set and predictions

This appendix supports §11.1.4 by defining a property class whose denotation is channel-relative, enabling a principled test of 'anchor collapse'.

L1. Property inventory

- Visual-only: blurry, hazy, pixelated, out-of-focus.
- Auditory-only: muffled, tinny, distorted.

- Cross-modal but channel-relative: faint, unclear, noisy.

L2. Minimal pairs (illustrative)

(i) The image is blurry. / The image looks blurry.
(ii) The audio is muffled. / The audio sounds muffled.
(iii) #The audio looks muffled. / #The image sounds blurry. (cross-channel mismatch)

L3. Predictions

- Reduced BE–LOOK contrast for visual-only properties when the evaluation channel is fixed by context.
- Sharper degradation for cross-channel mismatch than for ordinary △ cells in Table 7.
- A refined stability metric: Stable states remain well-defined under BE without specifying channel; channel-dependent states require channel anchoring or a contextually fixed measurement perspective.